Scientific Notation & Multiplying Monomials

2026-10-28

Rules to remember

Scientific: $a \times 10^n$, $1 \le a < 10$

Number ≥ 10: positive exponent

Number < 1: negative exponent

$a^m \cdot a^n = a^{m+n}$

Multiplying: multiply the numbers, then add the exponents

Example: multiply in scientific notation

$(4 \times 10^5)(3 \times 10^{-2})$

$= (4 \cdot 3) \times 10^{5+(-2)}$

$= 12 \times 10^3$

$= 1.2 \times 10^4$

Rule: multiply numbers, add exponents.

12 is not less than 10, so move the decimal.

Write in scientific notation.

1. 5,700

Rule: _____

Answer: _____

2. 906,000

Rule: _____

Answer: _____

3. 0.081

Rule: _____

Answer: _____

4. 0.0000043

Rule: _____

Answer: _____

Write in standard form.

5. 2.9×10^6

Rule: _____

Answer: _____

6. 1.004×10^4

Rule: _____

Answer: _____

7. 8×10^{-2}

Rule: _____

Answer: _____

8. 3.75×10^{-4}

Rule: _____

Answer: _____

Multiply. Simplify your answer.

9. $b^4 \cdot b^3$

Rule: _____

Answer: _____

10. $2x^5 \cdot x^4$

Rule: _____

Answer: _____

11. $4a^3 \cdot 6a^2$

Rule: _____

Answer: _____

12. $-3m^2 \cdot 5m^7$

Rule: _____

Answer: _____

13. $6n^4 \cdot (-2n^5)$

Rule: _____

Answer: _____

14. $-8k \cdot (-2k^3)$

Rule: _____

Answer: _____

Multiply. Write your answer in scientific notation.

15. $(2 \times 10^3)(4 \times 10^5)$

Rule: _____

Answer: _____

16. $(3 \times 10^4)(1.5 \times 10^2)$

Rule: _____

Answer: _____

17. $(2.5 \times 10^{-3})(3 \times 10^6)$

Rule: _____

Answer: _____

18. $(4 \times 10^{-2})(2 \times 10^{-5})$

Rule: _____

Answer: _____

19. $(5 \times 10^3)(6 \times 10^4)$

Rule: _____

Answer: _____

20. $(8 \times 10^{-4})(5 \times 10^{-2})$

Rule: _____

Answer: _____